

Cross-Level Residual Frequency-Guided Multi-Scale Refinement with a ConvNeXt Encoder for Polyp Segmentation

Abstract

Accurate colorectal polyp segmentation remains challenging because lesions vary in scale, morphology, contrast, and boundary appearance. Conventional multi-scale convolution blocks aggregate different receptive fields without explicitly using sample-specific frequency information from deep semantic features. We present a ConvNeXt-Tiny U-shaped framework with bottleneck Frequency-Aware Feature Enhancement (FAFEM) and introduce residual frequency-guided coupling to the Multi-Scale Convolution Block (MSCB) structure. The resulting RFG-MSCB reuses the low- and high-frequency descriptor produced by bottleneck FAFEM to assign sample-dependent weights to MSCB multi-kernel branches at the Stage-3 encoder-decoder fusion. The guided response is a bounded residual deviation from ordinary MSCB aggregation, preserving the conventional multi-scale path as a reference. The model is evaluated under a PraNet-style data protocol using final seen test sets from Kvasir-SEG and CVC-ClinicDB and external test sets from CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB. Checkpoint selection uses only an internal validation subset of the combined Kvasir-SEG/CVC-ClinicDB development pool. The architectural ablation changes mDice/mIoU/MAE from 0.827/0.763/0.019 for the Baseline to 0.869/0.803/0.018 for FAFEM + MSCB and 0.875/0.815/0.017 for Residual RFG-MSCB. The final modification adds 0.08 M parameters over FAFEM + MSCB without additional reported GFLOPs at the displayed precision, while preserving the conventional MSCB pathway.

Introduction

Colorectal polyps are abnormal tissue growths that can develop into malignant lesions if they are not detected and removed at an early stage. Colonoscopy is widely used for the examination of the lower gastrointestinal tract and enables clinicians to identify and remove suspicious lesions. Nevertheless, visual inspection remains challenging because polyp appearance varies considerably across patients and imaging conditions. Variations in lesion size, morphology, texture, contrast, and location, together with specular reflections, motion blur, intestinal folds, bubbles, and other visual artifacts, can complicate accurate delineation of polyp regions. Automated polyp segmentation therefore remains an important problem in computer-aided colonoscopy, where reliable pixel-level localization may assist subsequent analysis and clinical decision support.

Accurate polyp segmentation requires a model to simultaneously capture high-level semantic information and preserve sufficiently detailed spatial representations. U-shaped encoder-decoder architectures are well suited to this task because they combine progressively abstract encoder features with higher-resolution representations through skip connections. U-Net established this general design, while subsequent architectures such as UNet++ further demonstrated the importance of multi-level feature interaction and hierarchical decoding [1, 2]. However, repeated downsampling reduces spatial resolution, and fixed decoder operations may not always provide sufficient flexibility for lesions that vary considerably in scale and visual frequency content.

Recent transformer-based and hybrid architectures have improved long-range dependency modeling in medical image segmentation. TransUNet combines convolutional representations with transformer encoding, while Polyp-PVT applies hierarchical transformer features and cross-level aggregation to polyp segmentation [3, 4]. These approaches demonstrate the importance of broad contextual modeling, but they can introduce additional architectural complexity. Modern convolutional networks such as ConvNeXt provide an alternative by retaining the efficiency and spatial inductive bias of convolution while incorporating design principles that improve representation capacity. ConvNeXt employs large-kernel depthwise convolution, inverted channel

expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, making it a strong hierarchical encoder for dense prediction tasks [5].

Multi-scale representation is particularly important in polyp segmentation because lesions may occupy very small regions or extend over a large portion of the image. Multi-Scale Convolution Blocks (MSCBs) address this issue by combining convolutional branches with different receptive fields, allowing a network to aggregate local and broader spatial responses [6]. In their ordinary form, the branch responses are summed without sample-specific frequency conditioning.

Frequency-aware feature processing provides another complementary perspective. The bottleneck FAFEM implemented in this work uses local low-pass and residual high-pass proxies to enhance deep features and produce a compact descriptor. The descriptor can then provide image-dependent information for guiding later multi-scale processing rather than being used only at the bottleneck.

This observation motivates the central idea of this work. Instead of treating bottleneck frequency enhancement and decoder multi-scale refinement as independent operations, we establish an explicit cross-level interaction between them. The frequency descriptor produced by a bottleneck FAFEM is propagated to the first decoder stage and used to regulate the relative contributions of the multi-scale branches in an MSCB. In this way, the receptive-field mixture of the decoder becomes sample dependent rather than fixed across all images.

Directly replacing the original MSCB aggregation with a fully frequency-guided formulation could, however, unnecessarily perturb a stable multi-scale representation. To address this issue, we introduce the bounded residual guidance strategy formalized in Eq. (23). Here, M_{ord} denotes the ordinary multi-scale aggregation, M_{guide} denotes the frequency-conditioned aggregation, and s is a bounded learnable guidance-strength parameter. This formulation preserves the conventional MSCB pathway as a reference and allows the model to progressively learn frequency-dependent deviations during optimization.

Based on this mechanism, we develop a ConvNeXt-Tiny U-shaped segmentation framework in which the frequency-guided modification is localized to the first encoder–decoder fusion stage; the remaining decoder stages retain conventional fusion and depthwise-separable convolution.

The principal contributions of this work are summarized as follows:

- We introduce RFG-MSCB, which uses the bottleneck FAFEM descriptor to modulate the multi-kernel MSCB branches through bounded residual interpolation around ordinary MSCB aggregation.
- We evaluate the resulting ConvNeXt-Tiny segmentation architecture on multiple public colorectal polyp datasets using a three-seed comparison of FAFEM, FAFEM + MSCB, and FAFEM + RFG-MSCB configurations, supplemented by exploratory single-seed mechanistic and placement observations.
- We quantify the architectural trade-off: the localized RFG-MSCB modification adds 0.08 M parameters over FAFEM + MSCB without additional reported GFLOPs at the displayed precision.

Related work

Encoder–decoder architectures for medical image segmentation

Fully convolutional encoder–decoder networks remain a central design pattern for biomedical image segmentation. U-Net introduced a symmetric contracting and expanding architecture in which high-resolution encoder features are transferred to the decoder through skip connections, allowing semantic context and spatial detail to be combined during prediction [1]. UNet++ further showed that the structure of skip pathways is important by using nested dense connections to reduce the semantic discrepancy between encoder and decoder representations [2]. These works established the importance of hierarchical feature reuse, but they do not prescribe how the decoder should adapt its receptive-field mixture to the image-specific content of a polyp.

Polyp-specific convolutional models have explored efficient alternatives to large attention-based designs. HarDNet-MSEG combines a low-memory-traffic convolutional backbone with a lightweight decoder and demonstrates that strong polyp segmentation performance can be achieved without a transformer-heavy

architecture [7]. PraNet introduced a parallel partial decoder and reverse attention to progressively refine uncertain polyp regions [8]. These methods highlight the value of efficient decoding and iterative refinement, but their main mechanisms are not designed to use frequency information from the bottleneck to regulate multi-scale convolutional branches at an earlier decoder stage.

Modern convolutional encoders provide another route to high-capacity dense prediction. ConvNeXt revisits convolutional network design with large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, producing a hierarchical backbone with strong representation capacity [5]. We use ConvNeXt-Tiny as the encoder and focus the new mechanism on the interaction between bottleneck frequency information and Stage-3 multi-scale convolution.

Transformer and contextual polyp segmentation

The large variation in polyp size, shape, color, and boundary contrast motivates architectures that capture both local appearance and broader anatomical context. Transformer-based approaches address this need by modeling long-range spatial relationships. TransUNet combines convolutional feature extraction with a transformer encoder for medical image segmentation [3], and Polyp-PVT applies pyramid vision transformer features and specialized fusion to polyp segmentation [4]. Recent polyp-segmentation studies have continued this direction through contrastive transformer modeling, contextual information flow, phase-wise feature pyramids, and pyramid-transformer feature fusion [9, 10, 11, 12].

These approaches show the value of global context and cross-level interaction, while motivating a convolutional alternative to more complex attention-based designs.

Efficient multi-scale convolutional decoding

Multi-scale convolution is widely used because polyps may appear as small, subtle lesions or occupy a large image region. Parallel kernels and dilated or depthwise branches allow a decoder to aggregate local and broader responses without relying entirely on self-attention. EMCAD introduced an efficient multi-scale convolutional attention decoder for medical image segmentation and uses Multi-Scale Convolution Blocks (MSCBs) as part of its lightweight decoding design [6]. In an MSCB, multiple convolutional branches provide different receptive fields and are combined to form a multi-scale response.

MSCB provides the multi-scale reference path used by the present comparison.

Frequency-aware and adaptive feature recalibration

Attention and recalibration mechanisms provide a general precedent for using compact descriptors to control feature responses. Squeeze-and-Excitation Networks use global channel descriptors to recalibrate convolutional features [13], while Selective Kernel Networks adaptively weight convolutional branches with different receptive fields [14]. These methods establish descriptor-driven recalibration and adaptive receptive-field selection as relevant precedents.

Frequency-aware attention provides a closer conceptual link. FcaNet interprets channel attention through frequency analysis and uses frequency components for channel attention [15]. This line of work motivates frequency-aware recalibration, while the present work applies that idea to the relationship between bottleneck enhancement and decoder refinement.

RFG-MSCB differs from these precedents in where its descriptor is formed and how it is applied. SE recalibrates channels, Selective Kernel Networks weight receptive-field branches using a descriptor from the same feature level, and FcaNet constructs frequency-informed channel attention. RFG-MSCB instead forms a low/high spatial-frequency proxy at the bottleneck, transfers its descriptor across levels to the Stage-3 decoder fusion, and targets the aggregation of the MSCB multi-kernel branches. The ordinary MSCB sum remains explicit, and the guided response enters only through bounded residual interpolation. The resulting mechanism combines established descriptor-driven branch weighting with cross-level, bounded coupling to the MSCB reference path.

Boundary and recent polyp-segmentation refinements

Weak and visually ambiguous lesion boundaries remain a major challenge in polyp segmentation. Boundary-aware methods often introduce explicit edge supervision, auxiliary contour branches, image-pyramid high-frequency cues, or reverse attention. MEGANet employs multi-scale edge-guided attention for weak-boundary polyp segmentation [16], and MSBP-Net performs multi-scale boundary prediction to improve contour delineation [17]. Other recent methods model uncertainty, contextual flow, and multi-stage refinement [18, 10, 19].

Boundary awareness is introduced through the training objective rather than an explicit boundary-refinement branch.

Positioning of the method

Existing work has improved polyp segmentation through U-shaped skip reuse, reverse attention, transformer-based context modeling, efficient convolutional decoding, frequency-aware representation learning, and boundary-aware refinement. The method is positioned at the intersection of efficient multi-scale convolution and frequency-aware recalibration.

Specifically, the final model combines a ConvNeXt-Tiny encoder, a locally implemented bottleneck FAFEM, and multi-scale decoding based on the MSCB structure [6]. We introduce cross-level bounded residual modulation of the MSCB branches using the bottleneck descriptor, as developed technically in the next section.

Method

Overall Architecture

The network uses a U-shaped encoder–decoder architecture built upon a ConvNeXt-Tiny encoder. Frequency information extracted from deep semantic features regulates multi-scale processing during decoding. Frequency enhancement and multi-scale convolution are linked through a cross-level connection between the bottleneck and the first decoder stage.

Let $I \in \mathbb{R}^{3 \times H \times W}$ denote an input colonoscopy image. The ConvNeXt-Tiny encoder extracts four hierarchical feature maps,

$$\{F_1, F_2, F_3, F_4\} = E(I), \quad (1)$$

where $F_1 \in \mathbb{R}^{96 \times H/4 \times W/4}$, $F_2 \in \mathbb{R}^{192 \times H/8 \times W/8}$, $F_3 \in \mathbb{R}^{384 \times H/16 \times W/16}$, and $F_4 \in \mathbb{R}^{768 \times H/32 \times W/32}$. For an input resolution of 352×352 , their spatial dimensions are 88×88 , 44×44 , 22×22 , and 11×11 , respectively.

The deepest encoder feature F_4 is first processed by a Frequency-Aware Feature Enhancement Module (FAFEM), which produces an enhanced bottleneck representation together with a compact frequency descriptor. The enhanced representation is passed through a lightweight residual bottleneck, whereas the frequency descriptor is propagated to the first decoder stage. After the decoder feature is fused with F_3 , a Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses the frequency descriptor to adaptively regulate its multi-scale convolution branches.

The overall processing can therefore be summarized as

$$(\tilde{F}_4, \mathbf{z}_f) = \mathcal{F}_{\text{FAFEM}}(F_4), \quad (2)$$

$$B = \mathcal{B}(\tilde{F}_4), \quad (3)$$

$$D_4 = \mathcal{M}_{\text{RFG}}(\mathcal{S}_3[\mathcal{D}_4(B), F_3], \mathbf{z}_f). \quad (4)$$

followed by

$$D_3 = \mathcal{S}_2 [\mathcal{D}_3(D_4), F_2], \quad (5)$$

$$D_2 = \mathcal{S}_1 [\mathcal{D}_2(D_3), F_1], \quad (6)$$

$$D_1 = \mathcal{D}_1(D_2). \quad (7)$$

Here, $\mathcal{D}_i$ denotes a decoder block, $\mathcal{S}_i$ denotes the corresponding skip-feature fusion operation, $\mathbf{z}_f$ is the frequency descriptor extracted at the bottleneck, and $\mathcal{M}_{\text{RFG}}$ denotes the residual frequency-guided multi-scale refinement mechanism. Equations (4)–(7) summarize the decoder path.

The final prediction head transforms D_1 into a single-channel segmentation logit map, which is bilinearly resized to the original image resolution.

ConvNeXt-Tiny Encoder

The encoder employs ConvNeXt-Tiny with stage depths $[3, 3, 9, 3]$ and channel dimensions $[96, 192, 384, 768]$. Before entering the encoder, each RGB image is normalized using the ImageNet statistics $\boldsymbol{\mu} = (0.485, 0.456, 0.406)$ and $\boldsymbol{\sigma} = (0.229, 0.224, 0.225)$.

The first stage uses a 4×4 convolution with stride 4 to generate F_1 . Each subsequent stage begins with spatial downsampling using a 2×2 stride-2 convolution.

For an input feature X , a ConvNeXt block first performs a 7×7 depthwise convolution. The resulting feature is normalized in channels-last format and processed by a pointwise multilayer perceptron with a fourfold channel expansion:

$$\begin{aligned} Z &= \text{DWConv}_{7 \times 7}(X), \\ V &= W_2(\text{GELU}(W_1(\text{LN}(Z)))) \end{aligned} \quad (8)$$

The block output is obtained through layer-scaled residual learning,

$$Y = X + \text{DropPath}(\gamma \odot V), \quad (9)$$

where γ is a learnable layer-scale parameter initialized to 10^{-6} .

A stochastic-depth rate of 0.25 is used throughout the encoder. Feature dropout with probability 0.2 is additionally applied after the second, third, and fourth encoder stages.

Frequency-Aware Bottleneck Enhancement

The deepest encoder representation F_4 contains strong semantic information but has undergone substantial spatial downsampling. To expose complementary frequency characteristics of this representation, FAFEM is applied exclusively at the bottleneck.

Given $X = F_4$, a local low-frequency approximation is obtained using 3×3 average pooling with stride 1 and padding 1,

$$X_L = \text{AvgPool}_{3 \times 3}(X), \quad (10)$$

$$X_H = X - X_L. \quad (11)$$

The low- and high-frequency components in Eqs. (10) and (11) are independently transformed using depthwise convolutional branches,

$$\hat{X}_L = \phi_L(X_L), \quad \hat{X}_H = \phi_H(X_H), \quad (12)$$

where each $\phi(\cdot)$ contains a 3×3 depthwise convolution followed by GELU activation.

The two components are concatenated and globally pooled to obtain the frequency descriptor

$$\mathbf{z}_f = \text{GAP}([\hat{X}_L, \hat{X}_H]) \in \mathbb{R}^{2C}. \quad (13)$$

Since $C = 768$ at the bottleneck, $\mathbf{z}_f \in \mathbb{R}^{1536}$.

A lightweight channel-gating network applies global average pooling to $[\hat{X}_L, \hat{X}_H]$, followed by 1×1 transformations $1536 \rightarrow 48 \rightarrow 1536$, GELU after the first transformation, and a sigmoid after the second. Splitting the output channel-wise gives $w_L, w_H \in (0, 1)^{768}$; these weights are not normalized to sum to one. Their weighted combination is spatially refined and introduced through a residual connection:

$$\tilde{F}_4 = F_4 + \gamma_f \mathcal{R}_f(w_L \odot \hat{X}_L + w_H \odot \hat{X}_H), \quad (14)$$

where $\mathcal{R}_f$ is a bias-free 3×3 depthwise convolution with stride 1 and padding 1, and γ_f is a learnable scalar initialized to zero. The descriptor $\mathbf{z}_f$ is pooled before the channel weights are applied.

Lightweight Residual Bottleneck

The frequency-enhanced representation $\tilde{F}_4$ is subsequently processed by a lightweight residual bottleneck.

A 1×1 convolution first reduces the number of channels from 768 to 192:

$$X_r = \text{GELU}\left(\text{LN}\left(\text{Conv}_{1 \times 1}(\tilde{F}_4)\right)\right). \quad (15)$$

A bias-free 3×3 depthwise-separable convolution with padding 1 performs spatial transformation in the reduced feature space, followed by channels-first Layer Normalization ($\epsilon = 10^{-6}$), GELU, and spatial dropout with probability 0.2. A bias-free 1×1 convolution then projects the representation back to 768 channels and is followed by channels-first Layer Normalization.

The bottleneck output is formulated as

$$B = \tilde{F}_4 + \gamma_b \mathcal{H}(\tilde{F}_4), \quad (16)$$

where $\mathcal{H}(\cdot)$ denotes the bottleneck transformation and γ_b is a learnable scalar initialized to zero.

This initialization makes the bottleneck behave initially as an identity mapping and allows its transformation to be progressively introduced during optimization.

Residual Frequency-Guided Multi-Scale Convolution

The principal architectural contribution of this work is the Residual Frequency-Guided Multi-Scale Convolution mechanism.

After the bottleneck feature is upsampled from $H/32$ to $H/16$, the decoder representation is reduced from 768 to 384 channels and concatenated with the Stage-3 encoder feature F_3 . A lightweight fusion block transforms the resulting 768-channel representation into $X_3 \in \mathbb{R}^{384 \times H/16 \times W/16}$, or $\mathbb{R}^{384 \times 22 \times 22}$ for a 352×352 input.

A conventional lightweight multi-scale convolution block first expands the feature dimension from C to $2C$:

$$X_e = \text{ReLU6}(\text{BN}(\text{Conv}_{1 \times 1}(X_3))). \quad (17)$$

Three parallel depthwise convolution branches with kernel sizes $k \in \{1, 3, 5\}$ then generate multi-scale responses,

$$B_k = \phi_k(X_e), \quad (18)$$

where each ϕ_k consists of a depthwise convolution, Batch Normalization, and ReLU6 activation.

In a conventional MSCB, these branches are combined uniformly:

$$M_{\text{ord}} = B_1 + B_3 + B_5. \quad (19)$$

The new mechanism conditions the relative contributions of the three receptive fields on the bottleneck frequency descriptor $\mathbf{z}_f$.

A lightweight guidance network maps the 1536-dimensional descriptor to three branch logits:

$$\mathbf{q} = W_{g2}(\text{GELU}(W_{g1}\mathbf{z}_f)), \quad (20)$$

where $W_{g1} : 1536 \rightarrow 48$ and $W_{g2} : 48 \rightarrow 3$.

The normalized branch weights are calculated as

$$[\alpha_1, \alpha_3, \alpha_5] = 3 \text{Softmax}(\mathbf{q}), \quad (21)$$

such that $\alpha_1 + \alpha_3 + \alpha_5 = 3$.

The frequency-guided response becomes

$$M_{\text{guide}} = \alpha_1 B_1 + \alpha_3 B_3 + \alpha_5 B_5. \quad (22)$$

We introduce the guidance as a bounded residual deviation from the conventional multi-scale response:

$$M_{\text{RFG}} = M_{\text{ord}} + s(M_{\text{guide}} - M_{\text{ord}}), \quad (23)$$

where $0 \leq s \leq 0.5$.

The implementation stores an unconstrained learnable scalar θ_s , initialized to 0.05, and uses the hard-clamped effective strength $s = \text{clamp}(\theta_s, 0, 0.5)$ in Eq. (23).

The final guidance-layer weights and biases are initialized so that the branch logits are equal. Thus, at initialization, $\alpha_1 = \alpha_3 = \alpha_5 = 1$ and $M_{\text{guide}} = M_{\text{ord}}$.

The module thus initially reproduces the conventional multi-scale path, while optimization is allowed to learn image-dependent deviations from uniform branch weighting.

The resulting feature is processed using channel shuffle with $\text{gcd}(2C, C) = 384$ groups and a bias-free 1×1 projection from $2C$ back to C , followed by Batch Normalization. This sequence defines $\mathcal{P}$ below. A final residual connection preserves the input representation:

$$Y_3 = X_3 + \mathcal{P}(\text{Shuffle}(M_{\text{RFG}})). \quad (24)$$

This formulation connects the bottleneck frequency descriptor to multi-scale spatial refinement at $H/16$ resolution.

Progressive Decoder and Skip Fusion

After frequency-guided Stage-3 refinement, the decoder progressively restores spatial resolution using bilinear upsampling and depthwise-separable convolution.

The first refined decoder feature has 384 channels, $D_4 = Y_3$.

The subsequent decoder stages follow Eqs. (5)–(7).

For a 352×352 input, the decoder dimensions are $D_4 : 384 \times 22 \times 22$, $D_3 : 192 \times 44 \times 44$, $D_2 : 96 \times 88 \times 88$, and $D_1 : 96 \times 176 \times 176$.

At each skip level, the decoder feature is resized to the skip-feature shape when needed using bilinear interpolation with `align_corners=False`, concatenated with the corresponding encoder feature, and processed by a lightweight fusion block. The fusion block contains a bias-free 1×1 projection, channels-first Layer Normalization ($\epsilon = 10^{-6}$), GELU, a bias-free depthwise-separable 3×3 convolution with padding 1, a second channels-first Layer Normalization and GELU, and spatial dropout with probability 0.2. The primary twofold upsampling inside each decoder block uses bilinear interpolation with `align_corners=True`.

No additional attention mechanism, frequency module, MSCB, or boundary-specific refinement block is applied at these decoder stages.

The final decoder stage has no direct encoder skip connection because the ConvNeXt stem begins at one-quarter of the input resolution.

Prediction Head

The final decoder feature D_1 is processed using a lightweight refinement head. Two depthwise-separable 3×3 convolutions transform the feature dimensions as $96 \rightarrow 96 \rightarrow 48$.

Each transformation is followed by Layer Normalization and GELU activation. A final 1×1 convolution produces a single-channel segmentation logit map:

$$P_0 = \text{Conv}_{1 \times 1}(\mathcal{R}(D_1)), \quad (25)$$

$$P = \text{Up}_{H,W}(P_0), \quad (26)$$

$$\hat{Y} = \sigma(P). \quad (27)$$

The prediction in Eq. (26) is bilinearly resized from $H/2 \times W/2$ to the original input resolution with `align_corners=False`. The network directly returns logits; during inference, Eq. (27) gives the foreground probability map,

where $\sigma(\cdot)$ denotes the sigmoid function.

The selected architecture does not employ auxiliary prediction heads or deep supervision.

Model Complexity

The complete model has 30.06 M parameters and requires 27.82 GFLOPs for a 352×352 input. RFG-MSCB is localized to the Stage-3 encoder–decoder fusion, while the remaining decoder retains depthwise-separable convolution.

Experiments

Datasets and Data Splits

To evaluate the proposed framework under a fixed training and evaluation design, we used Kvasir-SEG [23], CVC-ClinicDB [24], CVC-300 [25], CVC-ColonDB [26], and ETIS-LaribPolypDB [27]. Following a

Table 1: Computational complexity comparison. Parameter counts and computational cost are taken from original publications or official implementations where available; ‘–’ denotes an unavailable or unverified value.

Method	Year	Input	Params (M)	GFLOPs
MEGANet (Res2Net-50) [16]	2024	352 ²	44.19	28.85
MSBP-Net [17]	2026	352 ²	25.52	12.86
CIFFormer [10]	2025	352 ²	32.44	13.14
PFPRNet [11]	2025	352 ²	118.52	39.39
MLB-Net [20]	2026	352 ²	87.70	–
MSPN [21]	2026	352 ²	28.56	13.21
DVFIP-Net [22]	2026	352 ²	45.62	19.37
Ours	–	352 ²	30.06	27.82 [†]

[†]Our value uses 13.91 GMACs and the explicit convention FLOPs = 2 × MACs. Competitor values are preserved in source terminology and are not convention-normalized.

PraNet-style data partition, 100 of the 1000 Kvasir-SEG images and 62 of the 612 CVC-ClinicDB images were reserved as final seen test sets. The remaining 900 Kvasir-SEG and 550 CVC-ClinicDB images formed a combined 1450-image train/validation pool. This pool was split deterministically into 1305 training images and 145 internal validation images using shuffled sampling with `random_state=42`; dataset membership was identical across all training seeds.

The final Kvasir-SEG and CVC-ClinicDB test images were excluded from training, validation, checkpoint selection, early stopping, threshold tuning, hyperparameter tuning, and architecture selection. CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB were used exclusively as external test sets and did not influence model selection.

Table 2: Dataset partition used in the experiments.

Dataset	Total	Train	Test
Kvasir-SEG	1000	900	100
CVC-ClinicDB	612	550	62
CVC-300	60	–	60
CVC-ColonDB	380	–	380
ETIS-LaribPolypDB	196	–	196

Images were converted to RGB, resized to 352×352 with bilinear interpolation, and scaled to $[0, 1]$. Masks were resized with nearest-neighbor interpolation and binarized. The encoder applied ImageNet normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

Implementation Details

The final protocol uses seeds 42, 6543, and 7777. All final and ablation results are reported as the mean $\pm$ sample standard deviation (ddof=1) across these seeds unless otherwise noted. The runs were implemented in PyTorch 2.5.1+cu124 on Windows, with a maximum of 200 epochs and batch size 24. AdamW with betas (0.9, 0.999) used separate parameter groups. The maximum learning rates were 3.0×10^{-4} for the decoder and 4.5×10^{-4} for the final refinement head. Encoder Stages 4, 3, 2, and 1 plus the stem used 3.0×10^{-5} , 2.4×10^{-5} , 1.92×10^{-5} , and 1.536×10^{-5} , respectively, corresponding to layer-wise decay 0.8. Weight decay was 1.0×10^{-4} for the decoder, 5.0×10^{-2} for the encoder, and 1.0×10^{-2} for the refinement head.

Learning rates were warmed up linearly for five epochs from 0.1 to 1.0 of each group maximum and then cosine-annealed over the remaining epochs to 1.0×10^{-6} . Training used AMP FP16, global gradient clipping at norm 1.0, and an exponential moving average (EMA) with decay 0.995. All parameters were trainable from the first epoch; no staged freezing or decoder-only warmup was used. The encoder was initialized from `convnext_tiny_22k_1k_384.pth`, logged as ImageNet pretrained weights. Deep supervision and frequency

augmentation were disabled, and sampling was uniform. Early stopping was configured with patience 30 from the start of training. Seeds 42 and 6543 completed the maximum 200 epochs, corresponding to zero-based epochs 0–199; seed 7777 stopped after epoch 167 (168 completed epochs). Model complexity and its measurement convention are reported in Table 1.

Training augmentation applied horizontal and vertical flips independently with probability 0.5, rotation in $[-15^\circ, 15^\circ]$ with probability 0.5, and a random crop ratio in $[0.7, 1.0]$ followed by resizing with probability 0.5. Joint brightness and contrast factors in $[0.85, 1.15]$ and saturation in $[0.9, 1.1]$ were applied with probability 0.5. A 3×3 Gaussian blur and additive Gaussian noise with standard deviation 0.015 were applied with probabilities 0.15 and 0.25, respectively.

The code seeded Python, NumPy, PyTorch CPU, and all CUDA random-number generators. Deterministic CUDA algorithms and cuDNN deterministic backend flags were not enabled.

Loss Function

The final objective was

$$\mathcal{L} = 0.55\mathcal{L}_{\text{Dice}} + 0.25\mathcal{L}_{\text{BCE}} + 0.20\mathcal{L}_{\text{boundary}}. \quad (28)$$

The objective in Eq. (28) uses a soft Dice term with a smoothing constant of 1.0. The BCE-with-logits term used targets smoothed as $y_{\text{BCE}} = 0.98y + 0.01$, corresponding to label smoothing 0.02. For the boundary-aware term, fixed Sobel filters were applied to each binary mask to obtain $g = \sqrt{g_x^2 + g_y^2}$. The magnitude was normalized per image as $a = g / \max(\max(g), 10^{-6})$, and the pixel weights were $w = 1 + \kappa a$ with $\kappa = 5$; consequently, the weights range from 1 to 6. For probability $p = \sigma(P)$ and target y , the term was

$$\mathcal{L}_{\text{boundary}} = -\frac{\sum_i w_i [y_i \log(p_i + \epsilon) + (1 - y_i) \log(1 - p_i + \epsilon)]}{\sum_i w_i + \epsilon} + 1 - \frac{\sum_i w_i y_i p_i + \epsilon}{\sum_i w_i (y_i + p_i - y_i p_i) + \epsilon}, \quad (29)$$

with $\epsilon = 10^{-6}$ and batch averaging for the weighted IoU component. The focal-Tversky term was assigned zero weight and therefore did not contribute to the final objective.

Evaluation Metrics

We report mean Dice (mDice), mean intersection over union (mIoU), S-measure S_α [28], weighted F-measure F_β^w [29], mean and maximum E-measure (mE_ϕ and $\max E_\phi$) [30], and mean absolute error (MAE) [31]. For each image, the thresholded prediction P and binary ground truth G were flattened and evaluated as

$$\text{Dice}(P, G) = \frac{2|P \cap G| + 10^{-6}}{|P| + |G| + 10^{-6}}, \quad \text{IoU}(P, G) = \frac{|P \cap G| + 10^{-6}}{|P| + |G| - |P \cap G| + 10^{-6}}. \quad (30)$$

The reported mDice and mIoU are arithmetic means of the per-image scores in Eq. (30), rather than global pixel aggregates. S-measure, weighted F-measure, the E-measure curve, and MAE were computed per image with `py_sod_metrics` from 8-bit probability maps and binary ground truth. The mean and maximum of the resulting E-measure curve were reported.

Model Selection and Evaluation Protocol

At each epoch, EMA weights were evaluated only on the 145-image internal validation subset. Probabilities were obtained with one sigmoid operation and binarized using the strict rule $p > 0.45$. A checkpoint was saved only when internal-validation IoU was strictly greater than the previous best. Best checkpoints were selected independently for each seed according to internal-validation IoU.

After model selection, each best checkpoint was loaded strictly and evaluated on the 100-image Kvasir-SEG and 62-image CVC-ClinicDB seen test sets, followed by the full CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB external test sets. Test-time augmentation was disabled. Neither the seen nor external test results influenced checkpoint selection or other model-development decisions.

Experimental Study

The experimental study first examines the contribution of the architectural components under controlled variants and then compares the resulting model with published methods. Presenting the ablation before the comparative evaluation establishes the reference configuration and clarifies which part of the final design is being assessed in the comparative tables.

Ablation Study

The ablation design compares three configurations in sequence: Baseline, FAFEM + MSCB, and Residual RFG-MSCB (Ours). This sequence evaluates the feature-enhancement and multi-scale components together before assessing the additional residual frequency-guided modification.

Ablation Results

The Baseline provides the reference without the added FAFEM and MSCB components.

Adding FAFEM and MSCB improves the baseline across the reported segmentation metrics. The final configuration then adds residual frequency-guided coupling to the multi-scale configuration. The complete numerical comparison, including computational complexity, is given in Table 3.

A single-seed direct frequency-guidance configuration tested the mechanism without residual coupling. Its Kvasir-SEG result was lower than that of FAFEM + MSCB in the paired seed-42 comparison.

Table 3: Ablation study of the proposed architectural components on Kvasir-SEG, including segmentation performance and computational complexity.

Model	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE	Params (M)	GFLOPs
Baseline	0.827	0.763	0.811	0.890	0.907	0.913	0.019	29.19	27.202
FAFEM + MSCB	0.869	0.803	0.845	0.910	0.937	0.947	0.018	29.98	27.818
Residual RFG-MSCB (Ours)	0.875	0.815	0.847	0.913	0.941	0.949	0.017	30.06	27.818

Ablation Interpretation

FAFEM + MSCB improves mDice from 0.827 to 0.869, mIoU from 0.763 to 0.803, and reduces MAE from 0.019 to 0.018 relative to the Baseline. Residual RFG-MSCB further changes mDice from 0.869 to 0.875, mIoU from 0.803 to 0.815, and MAE from 0.018 to 0.017. The parameter count increases from 29.19 M to 29.98 M and then 30.06 M, while GFLOPs increase from 27.202 to 27.818 and remain 27.818 for the final configuration. Thus, the residual modification adds only 0.08 M parameters over FAFEM + MSCB and no additional reported GFLOPs at the displayed precision. These are descriptive comparisons; no statistical significance is claimed.

Stage Placement and Generalization

Stage 3 is used in the selected configuration. The exploratory all-skip experiment used a single seed, was incomplete, and used batch size 32 with the same learning rate. Its training result was lower than that of the Stage-3 configuration. A controlled multi-seed placement study remains future work.

On the four additional datasets, the change from FAFEM + MSCB to the final guidance configuration was mixed in the paired seed-42 comparison: mDice decreased on CVC-ClinicDB and CVC-300 and increased on CVC-ColonDB and ETIS. The residual-coupling configuration with default initialization did not complete training and is therefore excluded from the quantitative ablation; initialization sensitivity remains a subject for future work.

Comparative study

After the component-level analysis, the comparative study evaluates the complete model on the Kvasir-SEG and CVC-ClinicDB seen test sets and on three external test sets, and places its performance alongside published baselines. The quantitative results are presented below.

Quantitative analysis

On Kvasir-SEG, the proposed model achieved the highest displayed mDice and mIoU, reaching 0.931 ± 0.004 and 0.883 ± 0.006 , respectively. The closest displayed method, PFPRNet, reported mDice of 0.930 and mIoU of 0.881. On CVC-ClinicDB, the proposed model achieved 0.937 ± 0.002 mDice and 0.888 ± 0.004 mIoU. PFPRNet reported higher values for these two metrics, while the proposed model remained competitive across the displayed comparison.

The 100 Kvasir-SEG and 62 CVC-ClinicDB images were reserved exclusively for final seen-set testing and did not participate in training, validation, checkpoint selection, or hyperparameter tuning. Checkpoint selection was performed exclusively on the 145-image internal validation subset sampled from the combined 1450-image Kvasir-SEG/CVC-ClinicDB train/validation pool.

The main comparison is restricted to methods using comparable PraNet-style data protocols with closely matched Kvasir-SEG and CVC-ClinicDB partitioning. Methods using materially different data partitions remain discussed in Related Work but are omitted from the main performance tables. Differences in optimization, augmentation, backbone initialization, model-selection criteria, and implementation details remain. Across the comparison tables, the best and second-best displayed values are shown in bold and underlined, respectively.

On the external datasets, mDice/mIoU were $0.882 \pm 0.002 / 0.810 \pm 0.002$ on CVC-300, $0.794 \pm 0.009 / 0.714 \pm 0.009$ on CVC-ColonDB, and $0.787 \pm 0.011 / 0.707 \pm 0.012$ on ETIS-LaribPolypDB. The largest between-seed variation in mDice and mIoU occurred on ETIS.

Table 4: Quantitative comparison on Kvasir-SEG.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MEGANet [16]	2024	0.913	0.863	0.907	0.918	0.956	0.959	0.025
MSBP-Net [17]	2026	0.919	0.868	0.916	0.926	0.961	–	0.023
CIFFormer [10]	2025	0.926	0.876	0.929	0.939	0.967	–	0.019
PFPRNet [11]	2025	<u>0.930</u>	<u>0.881</u>	0.920	0.935	0.971	–	0.019
MLB-Net [20]	2026	0.926	0.878	<u>0.921</u>	<u>0.938</u>	<u>0.969</u>	0.975	<u>0.021</u>
MSPN [21]	2026	0.927	0.879	–	–	–	–	<u>0.021</u>
DVFIP-Net [22]	2026	0.922	0.874	–	–	–	–	–
Ours	–	0.931	0.883	0.911	0.933	0.960	0.967	0.024
		± 0.004	± 0.006	± 0.005	± 0.003	± 0.003	± 0.003	± 0.001

Table 5: Quantitative comparison on CVC-ClinicDB.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ϕ}	$\max E_{\phi}$	MAE
MEGANet [16]	2024	0.938	0.894	0.940	0.950	–	<u>0.987</u>	<u>0.007</u>
MSBP-Net [17]	2026	0.940	0.892	0.927	0.939	0.976	–	<u>0.007</u>
CIFFormer [10]	2025	0.944	0.897	0.941	0.934	0.972	–	<u>0.007</u>
PFPRNet [11]	2025	0.949	0.903	<u>0.947</u>	<u>0.957</u>	0.989	–	0.006
MLB-Net [20]	2026	<u>0.945</u>	<u>0.902</u>	0.948	0.958	0.989	0.992	0.006
MSPN [21]	2026	0.943	<u>0.902</u>	–	–	–	–	0.006
DVFIP-Net [22]	2026	0.942	0.896	–	–	–	–	–

Continued on next page

Table 5: Quantitative comparison on CVC-ClinicDB (continued).

Methods	Year	mDice	mIoU	F_β^w	S_α	mE_ϕ	$\max E_\phi$	MAE
Ours	–	0.937 ± 0.002	0.888 ± 0.004	0.925 ± 0.003	0.947 ± 0.001	0.978 <u>± 0.001</u>	0.983 ± 0.001	0.011 ± 0.0003

Table 6: Quantitative comparison on ETIS-LaribPolypDB.

Methods	Year	mDice	mIoU	F_β^w	S_α	mE_ϕ	$\max E_\phi$	MAE
MEGANet [16]	2024	0.789	0.709	0.753	0.866	0.855	0.915	0.015
MSBP-Net [17]	2026	0.795	0.718	0.770	0.873	0.899	–	0.019
CIFFormer [10]	2025	0.810	<u>0.746</u>	0.858	0.889	<u>0.918</u>	–	<u>0.013</u>
PFPRNet [11]	2025	0.828	0.764	<u>0.839</u>	0.908	0.935	–	0.012
MLB-Net [20]	2026	<u>0.821</u>	0.744	0.780	<u>0.899</u>	0.914	0.935	<u>0.013</u>
MSPN [21]	2026	0.798	0.717	–	–	–	–	0.017
DVFIP-Net [22]	2026	0.811	0.730	–	–	–	–	–
Ours	–	0.787 ± 0.011	0.707 ± 0.012	0.753 ± 0.007	0.878 ± 0.004	0.902 ± 0.002	0.916 <u>± 0.003</u>	0.015 ± 0.001

Table 7: Quantitative comparison on CVC-ColonDB.

Methods	Year	mDice	mIoU	F_β^w	S_α	mE_ϕ	$\max E_\phi$	MAE
MEGANet [16]	2024	0.793	0.714	0.779	0.854	0.892	0.895	0.034
MSBP-Net [17]	2026	0.810	0.731	0.810	0.861	0.912	–	0.029
CIFFormer [10]	2025	0.823	0.741	0.851	0.874	<u>0.918</u>	–	<u>0.026</u>
PFPRNet [11]	2025	0.819	0.741	<u>0.843</u>	<u>0.882</u>	0.915	–	0.028
MLB-Net [20]	2026	0.832	0.757	0.825	0.886	0.924	0.931	0.023
MSPN [21]	2026	0.811	0.734	–	–	–	–	0.029
DVFIP-Net [22]	2026	<u>0.826</u>	<u>0.746</u>	–	–	–	–	–
Ours	–	0.794 ± 0.009	0.714 ± 0.009	0.772 ± 0.007	0.861 ± 0.006	0.898 ± 0.004	0.905 <u>± 0.004</u>	0.035 ± 0.001

Table 8: Quantitative comparison on CVC-300 (EndoScene).

Methods	Year	mDice	mIoU	F_β^w	S_α	mE_ϕ	$\max E_\phi$	MAE
MEGANet [16]	2024	0.899	0.834	0.882	0.935	0.966	0.969	0.007
MSBP-Net [17]	2026	0.903	0.859	0.927	<u>0.942</u>	<u>0.967</u>	–	0.019
CIFFormer [10]	2025	0.883	0.822	0.912	0.939	0.961	–	0.100
PFPRNet [11]	2025	0.891	0.833	<u>0.917</u>	0.928	0.950	–	0.013
MLB-Net [20]	2026	0.915	<u>0.852</u>	0.909	0.974	0.974	0.978	0.005
MSPN [21]	2026	<u>0.907</u>	0.840	–	–	–	–	<u>0.006</u>
DVFIP-Net [22]	2026	0.900	0.833	–	–	–	–	–
Ours	–	0.882 ± 0.002	0.810 ± 0.002	0.857 ± 0.001	0.928 ± 0.003	0.956 ± 0.005	0.971 <u>± 0.002</u>	0.007 ± 0.0003

Discussion

Quantitative and Ablation Interpretation

The evaluation protocol separates seen-domain testing on Kvasir-SEG and CVC-ClinicDB from external testing on CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB. All five test sets are isolated from checkpoint selection, which uses only the internal validation subset of the combined development pool. This design evaluates the sample-dependent guidance while retaining a convolutional multi-scale pathway for efficient decoding.

FAFEM establishes the bottleneck feature-enhancement reference, MSCB-lite adds multi-scale refinement at Stage 3, and RFG-MSCB adds adaptive branch modulation to that path. The final increment is small, consistent with a localized refinement of the decoder. In the paired single-seed comparison, direct frequency guidance without residual coupling produced a lower result than MSCB-lite.

The bounded residual design preserves the conventional MSCB response while allowing sample-dependent branch modulation. Because residualization and frequency conditioning were not evaluated independently, their individual contributions remain unresolved. This adaptive refinement reuses the bottleneck descriptor at one decoder stage without discarding the established multi-scale response or modifying the full decoder.

Cross-Dataset Generalization

CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB are reserved for external generalization evaluation and do not influence model development. Differences in acquisition, annotation, case mix, and polyp morphology make these datasets a distinct test of transfer beyond the Kvasir-SEG/CVC-ClinicDB development domain.

Although the main comparison is restricted to methods with closely matched dataset partitioning, differences in optimization, augmentation, backbone initialization, model-selection criteria, and implementation details remain.

Practical Trade-offs and Placement

As summarized in Table 1, the final architecture has 30.06 M parameters and requires 27.82 GFLOPs at 352×352 . Matched-hardware latency was not measured.

The final model uses the new guidance at the Stage-3 decoder–encoder fusion. The all-skip experiment was exploratory, single-seed, incomplete, and used a different batch size. Deployment-oriented placement and latency optimization remain future work.

Limitations

- **No statistical significance testing:** All reported differences are descriptive; no hypothesis testing was performed.
- **Hardware and software records:** Exact hardware and dependency versions were not recorded.
- **Training duration:** With zero-based epoch indices, seed 7777 stopped after epoch 167 (168 completed epochs; patience 30), while seeds 42 and 6543 completed epochs 0–199; this asymmetry may affect the mean estimate.

Future Work

- Systematic multi-seed placement ablation with controlled batch size and learning rate.
- Disentanglement of the RFG-MSCB components (residual form, initialization schedule, frequency conditioning) via factorial experiments.
- Investigation of initialization sensitivity using completed default-initialization runs.
- Domain adaptation or targeted augmentation to improve generalization on CVC-ColonDB and ETIS-LaribPolypDB.
- Evaluate between-configuration differences using larger seed ensembles or bootstrap confidence intervals.
- Record hardware and software configurations systematically in future experiments.

Conclusion

We presented a ConvNeXt-Tiny U-shaped segmentation framework that uses a locally implemented bottleneck FAFEM and couples its descriptor to the MSCB structure through RFG-MSCB at the Stage-3 encoder-decoder fusion. The design extends bottleneck frequency enhancement into decoder refinement while preserving the ordinary multi-scale pathway as a reference, thereby addressing the motivation for sample-dependent receptive-field selection without replacing the established decoder path.

The evaluation protocol reserves Kvasir-SEG and CVC-ClinicDB subsets for final seen-domain testing and uses CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB only for external testing; checkpoint selection is confined to the internal validation split. The parameter-efficient modification is localized to Stage 3, adds 0.08 M parameters over FAFEM + MSCB without additional reported GFLOPs at the displayed precision, and retains depthwise-separable convolution in the remaining decoder.

Future work should examine placement across multiple seeds, separate residualization from frequency conditioning, and evaluate domain-adaptation strategies for the external datasets. Overall, RFG-MSCB connects bottleneck frequency information to sample-dependent multi-scale decoding through bounded residual coupling that preserves the original MSCB pathway.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are publicly available from the sources cited in the manuscript. The project code, trained weights, and derived evaluation artifacts are maintained locally and are not currently available from a public repository.

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